



Mysterious 'Puffy' celestial objects

Strange, 'puffy' objects resembling clouds that act like stars, have been discovered near the Milky Way's supermassive black hole. <https://digit.in/jul18-73-1>



Cosmonauts play football

Two cosmonauts aboard the International Space Station celebrated the Football World Cup 2018 by playing soccer in zero-gravity. <https://digit.in/jul18-73-2>

sperm bilaterally, or unilaterally, even in the same species.

Though there are species of sexually differentiated snails, many of them are hermaphrodites. The hermaphrodite species have pointed structures of calcium and chitin known as "love darts". These are for sexual selection, with larger love darts increasing the chances of finding a mate. During the mating, the snails anchor themselves to their partners using the love darts. This keeps them together while the mating occurs. However, there is chance that the internal organs of the snails can be damaged because of the darts. The mating process can involve either bilateral or unilateral sperm exchange of sperm.

The clown fish (the orange and white fish from *Finding Nemo*) are an example of sequential hermaphrodites. All the clownfish are born males. They live in schools with many males and a single dominant female. The female lays eggs first, which is then fertilised by biggest males. When the dominant female dies, the largest male feeds voraciously, and turns into a female. Then the cycle continues.

PARASITIC MALES

A number of species show parasitic mating, where the male is permanently attached to the female, with providing sperm being the only function. These kind of species show extreme sexual dimorphism, where the males are generally much smaller than the female. Unless you know exactly where to look, the male might not even be visible. At other times, the males are absorbed into the body of the females.

The most well known example of this is the Anglerfish, some species are also known as Deep Sea Devils. The males in these species have only one function in life – to find a female to mate with. They cannot hunt food or even swallow, and all their sense organs - large eyes and extra sensitive noses, are only suitable for finding a female. If they cannot find a mate, they die. Once the males find a female, they bite into her skin, and release

an enzyme. This enzyme helps them digest the outer skin of the female, and their own mouths. After this process, the male is permanently fused to the side of the female, sharing a common circulatory system which provides the nutrients. Encounters with the same species in the deep sea is relatively rare, which is why these species have adopted this particular strategy for reproduction. In some species, the body of the male deteriorates with only the gonads left functional. Some Anglerfish species have only one parasitic partner, while females in other species may host up to eight parasitic males at the same time.

The *Osedax*, also known by the very cool name of zombie worms, were discovered as recently as 2002. They get their name because they feed on the bones of whale carcasses. There are entire harems of tiny males enclosed within gelatinous tubes of the females. Each female can harbour hundreds of males. The reproduction is continuous, giving rise to a large number of species in the genus. The *Osedax* might have been using this strategy for millions of years, considering the marks associated with the genus found on the whales of bones have also been found on plesiosaur fossils.

The *Bonellia viridis* or the green spoonworm is another example of parasitic dwarfism. The marine worms start off life as primitive organisms without a gender. If a larva touches the sea floor, it turns into a female and releases a toxin known as bonellin. The larvae that subsequently come into contact with the toxin, all turn into males. The males are then consumed by the female through a feeding tube. They spend the remainder of their lives inside the genital sac, providing the sperm as required.

THE DETACHABLE PENIS

The argonaut, or the paper nautilus is a kind of octopus that swims through jet propulsion and has a shell. There is extreme sexual dimorphism seen in these animals, with the females having bodies as large as 10 cm, and

shells as large as 30 cm. The males do not grow bigger than 2 cm. The males have a special arm, known as the hectocotylus, which contains the sperm. It detaches from the body of the male, swims by itself to the female, and deposits itself into her pallial sac, in the egg chamber. It resides there and provides the sperm needed to fertilise the eggs. A single female can host a hectocotylus from multiple males. After dispatching the "arm", the male argonaut dies. However, the female can continue to reproduce multiple times.

A species of sea slug known as the *Chromodoris reticulata*, has inversed the strategy adopted by the banana slugs. The species is made up of hermaphrodites, and after mating, they discard their own penises. These creatures cannot mate for another 24 hours, after which, they have a new penis. Coiled up within the body, are two stand by penises to take the place of the ditched organ. Researchers believe the sea slugs do this to prevent accidentally inseminating a mate with the sperm of a rival. There are backwards pointing spines on the penis, that can trap the sperm of a rival. There is a potential for these sperm to fertilise the egg on a subsequent mating, which is why the particular species has found it viable to cast off its penis.

Nephilengys malabarensis, also known as the orb-web spider is a species that displays sexual cannibalism. The female eats up the male after mating. During the mating, the male spider deposits its palp, or spider penis, into the genital opening of the female. The palp continues to inseminate the female even after being detached. It also acts as a plug, preventing other rival males from impregnating the female spider. If he survives the attempt to be eaten, the male remains in the vicinity, and continues to fend off rivals by fighting them. With its palp gone, the "eunuch males" are lighter, faster and more energetic than their rivals, who still have their palps with them. Studies have demonstrated that the eunuch spiders stand a better chance at winning a fight against rivals. 📌